

1.) Compare databases, data warehouses and data lakes.

Database	Data warehouse	Data lake
• Stores and manages current operational data. • Mainly structured data • Used for daily transactions, operations.	• Stores integrated historical data. • Mostly structured. • Used for Business intelligence, reporting and decision making.	• Stores large amounts of raw data. • Structured, semi-structured and unstructured. • Big-data analytics, exploration and machine learning.

2.) Galle municipal Council is interested in building a data warehouse to analyze traffic patterns within the city's road network and reduce congestion during rush hours. Briefly describe at least two (02) data sources that can assist them in analyzing these patterns and developing strategies. Clearly define what to extract?

* Data Source :- Traffic sensors.

* Extract :- Number of vehicles, Road/Junction location, Peak and off-peak times.

- Road sensors can provide information about number of vehicles using particular roads.
- Data is collected using road sensors at roads and junctions.
- This helps identify busy roads and improve traffic management.

* Data source :- Road accident data.

- Data collection :- Data is collected from police reports and hospital records.
- Extract :- Accident location, date, time, vehicles involved and accident frequency.
- This helps identify dangerous roads and improve road safety.

s) A tour operator offering trips to Ella in Sri Lanka is interested in understanding visitor preferences and formulating strategies to boost off-peak bookings. Briefly describe at least two data (or) sources that can assist the operator in analyzing visitor preferences and creating these strategies. Clearly define what to extract.

* Data source:- Hotel booking data.

- Data collection:- Data is collected from hotels and guesthouses in Ella.
- Extract:- Number of visitors, booking dates, length of stay, Nationality.
- This helps understand visitor patterns.

* Data source:- Customer feedback data

- Data collection:- Data is collected from feedback forms and reviews.
- Extract:- Favourite activities, ratings, complaints and customer preferences.
- This helps create better tour packages.